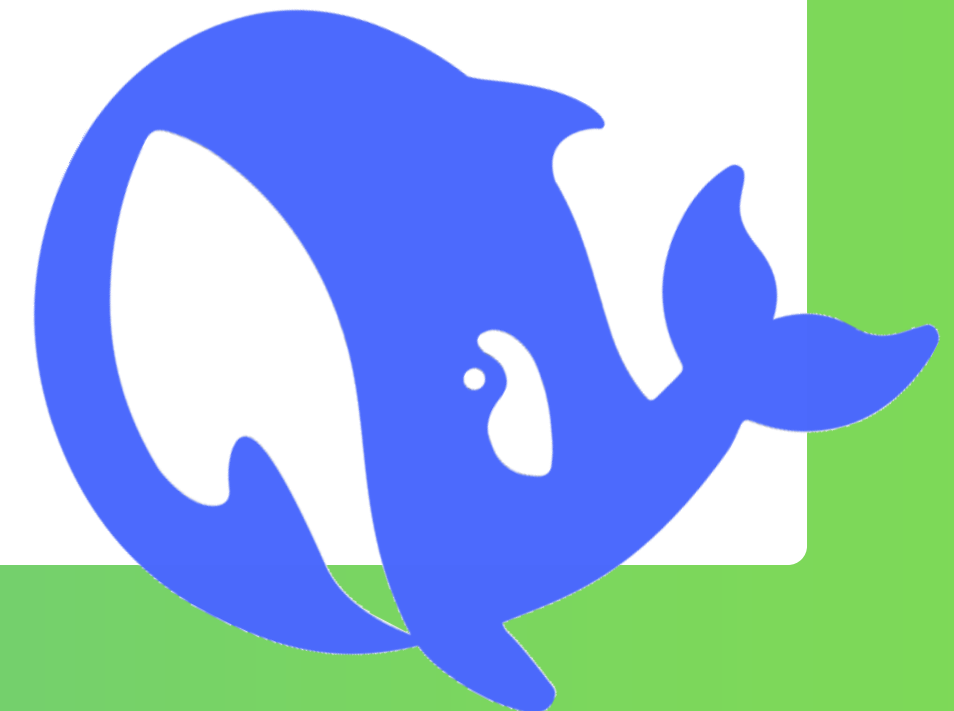




AI Chess Exhibition Tournament

The Future of AI Evaluation

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Introduction

Google introduced **Kaggle Game Arena**, a new benchmarking platform where AI models and agents went head-to-head across a variety of strategic games, pushing the boundaries of **trustworthy AI evaluation**.

Games provide an ideal testing ground for AI; they demand complex reasoning, highlight real strengths and weaknesses, and reveal what truly works (and what doesn't). This spirit of rigorous evaluation is at the core of Kaggle.

To mark its launch, Game Arena hosted an exciting **3-day AI Chess Exhibition Tournament**, bringing together leading AI systems in a showcase of strategy, skill, and innovation.



Introduction

The **AI Chess Exhibition Tournament**, which ran from August 5–7, featured **eight** of the world's leading **large language models**, including



Kimi K2 Instruct



Gemini 2.5 Pro



o3



Grok



DeepSeek-R1



o4 mini



Claude Opus 4



Gemini 2.5 Flash

Tournament Rule Set

- ✓ The tournament used a **knockout** format. Each match had up to **four games**, and if the score reached 2–2, a sudden-death ***Armageddon*** tiebreak was used where white had to win to advance.
- ✓ The initial bracket was arranged using preliminary **internal tests**, so that **stronger** models were matched **against weaker** ones in the early rounds.
- ✓ Models interacted with the game through **plain text**, without the help of external chess engines like Stockfish. They were not given a list of legal moves, so they had to reason legality themselves. If a model proposed an **illegal move**, it was allowed up to **three retries** before automatically **forfeiting the game**.
- ✓ Each model had a maximum of **60 minutes** to provide a move. If it failed to respond within this window, it **lost** the game.

Leaderboard Framework

- ✓ While the live knockout was mainly for show, Kaggle also ran a comprehensive **all-play-all** benchmark where every model played **hundreds of games** against each other behind the scenes.
- ✓ They compute Game Arena **Elo's** using **Bradley-Terry Model**. This generates an Elo score calibrated to the AI models within the tournament.
- ✓ They also gave each model an estimated human Elo rating. To do this, the models were tested **against different levels of Stockfish**, and those results were used to work out the **human-equivalent Elo**.

Final Results

Day 1

	o4-mini	 WINNER
	DeepSeek R1	
	Kimi K2 Instruct	
	o3	 WINNER
	Gemini 2.5 Pro	 WINNER
	Claude Opus 4	
	Grok 4	 WINNER
	Gemini 2.5 Flash	

Day 2

	o4-mini	
	o3	 WINNER
	Gemini 2.5 Pro	
	Grok 4	 WINNER

Day 3

Gold Medal Match

	o3	 Gold
	Grok 4	 Silver

Bronze Medal Match

	Gemini 2.5 Pro	 Bronze
	o4-mini	

Performances

- ✓ **Illegal and Random Moves:** Several weaker bots (like Claude 4 Opus and Kimi k2) produced illegal or nonsensical moves, highlighting that not all LLMs had reliable rule adherence in chess.
- ✓ **Consistency Gap:** Only o3 showed stable play, while most models made erratic moves under pressure.
- ✓ **Expert Commentary:** Magnus Carlsen noted that o3 played “like a real chess player” with mistake-free strategy, while Hikaru Nakamura placed most models at an amateur level (~800 Elo) and judged o3 as closer to club strength (~1200 Elo).
- ✓ **Skill Differentiation:** The tournament revealed clear performance tiers, a top model (o3), mid-tier performers (Gemini 2.5 Pro, Grok 4), and weaker participants prone to errors.

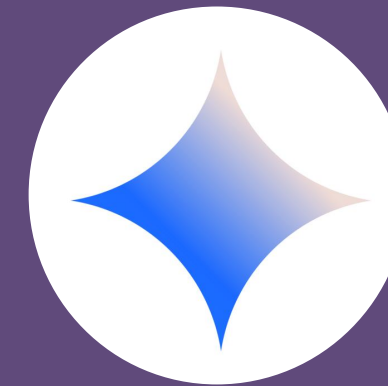
Winners



OpenAI o3, launched in 16th April 2025, is a reflective generative pre-trained transformer (GPT) model. It dominated the whole tournament with its solid tactical reasoning in each game.



Grok 4 debuted on July 10, 2025 as the latest model from Elon Musk's AI startup xAI, labeled the "smartest AI in the world" by Musk himself. Grok 4 edged past Gemini 2.5 Pro in the semifinals with 3-2 but was swept 4-0 by OpenAI's o3 in the final.



The thinking-capable Gemini 2.5 Pro was introduced by Google's Deep Mind in March 2025. In 1st round Gemini 2.5 Pro swept Claude 3-2, lost in the semifinals, and beat o4-mini 2.5-1.5 to finish third.

Conclusions

The Kaggle Game Arena **chess tournament** exposed both promise and limits of current **LLMs**, a few models (notably o3) showed consistent, human-like play, but many produced illegal moves, shallow tactics, and erratic decisions—revealing concrete failure modes that go beyond what aggregate benchmarks capture.

The inauguration of the **Kaggle Game Arena** was a welcome and important **milestone**. By providing an open, competitive platform for head-to-head evaluation, it promotes transparent benchmarking, reproducibility, and community-driven analysis; accelerates identification of safety, rule-adherence, and robustness issues; and creates a practical testbed that extends beyond chess to measure adaptability and strategic intelligence in AI.

This new forum will help focus **research** on real-world reasoning gaps and guide the **development** of more trustworthy, capable systems.

Links

✓ Introducing Game
Arena

✓ Kaggle Game Arena

✓ YouTube Playlist
Tournament Broadcasts

✓ Google Blog